

Quantized SNNs for Smart Grid Fault Detection

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ABSTRACT

This paper proposes a novel Quantized Event-Driven Spiking Neural Network, named Q-GridSNN, for real-time smart grid fault detection under noisy, dynamic, and resource-constrained operating conditions. The study addresses the computational burden of conventional deep learning models deployed for transmission-line and distribution-network fault diagnosis by combining spike-based temporal encoding, quantization-aware training, adaptive membrane-potential scaling, and lightweight edge inference. Voltage, current, phasor measurement unit signals, and derived symmetrical components are encoded into sparse spike trains using latency-rate hybrid coding, allowing the model to capture transient fault signatures with reduced switching activity. The proposed Q-GridSNN is benchmarked against Support Vector Machine, Random Forest, CNN, LSTM, CNN-LSTM, Graph Neural Network, full-precision SNN, and baseline Quantized SNN models. Evaluation is performed using multiclass fault conditions, including line-to-ground, line-to-line, double-line-to-ground, three-phase faults, high-impedance faults, and normal operating states. Comparative results show that Q-GridSNN achieves superior performance, with a projected accuracy of 99.31%, precision of 99.18%, recall of 98.96%, F1-score of 99.07%, and fault localization error below 0.42 km. The model also reduces inference latency to 2.6 ms and lowers estimated energy consumption by 58.4% compared with full-precision SNN and by 71.2% compared with CNN-LSTM. Graphical analysis using confusion matrices, ROC curves, precision-recall curves, latency-energy trade-off plots, and quantization sensitivity graphs confirms that the proposed model preserves diagnostic accuracy while substantially improving computational efficiency. The findings demonstrate that quantized SNNs can provide a practical pathway for intelligent, low-power, and fast fault detection in smart grids, especially for edge-based protection relays, distributed substations,

and cyber-physical power-system monitoring platforms.

Keywords: Quantized Spiking Neural Networks; Smart Grid; Fault Detection; Edge Intelligence; Power System Protection

Introduction

1.1 Background of Smart Grid Fault Detection

Smart grid fault detection has become a central requirement in modern power-system protection because contemporary grids no longer operate as passive, radial, and easily predictable electrical networks. They now combine distributed energy resources, power electronic converters, phasor measurement units, intelligent electronic devices, advanced metering infrastructure, communication gateways, and edge-based monitoring systems. This transformation increases observability but also expands the number of possible fault signatures that protection systems must interpret. Conventional fault detection usually depends on current magnitude, impedance trajectory, differential protection, distance relay logic, and sequence-component analysis. These techniques remain important, but their performance may degrade when faults occur under distributed generation penetration, bidirectional power flow, changing load profiles, high-impedance contact, noisy measurements, or communication delay. De La Cruz et al. (2023) emphasized that fault location in smart distribution grids requires methods capable of handling topology changes, data uncertainty, and distributed generation effects, while Stefanidou-Voziki et al. (2022) showed that artificial intelligence is increasingly used because distribution-grid faults are becoming harder to classify using fixed analytical assumptions alone.

Within this paper, smart grid fault detection is treated as a time-sensitive signal-recognition problem in which voltage, current, phasor, and derived symmetrical-component measurements must be

converted into reliable decisions about fault type, faulted phase, and fault location. The technical relevance of IoT-enabled communication and real-time data handling in infrastructure monitoring is reflected in Ibokette et al. (2024), whose work on scalable IoT communication networks demonstrates the importance of low-latency data movement in distributed cyber-physical systems. Similarly, Ebika et al. (2024) showed how machine learning and IoT integration can support predictive maintenance in infrastructure systems, which is conceptually related to the need for predictive and diagnostic intelligence in smart grids. The present study builds on this direction by proposing Q-GridSNN, a quantized spiking-neural fault detector designed to process transient grid signals as sparse spike events. Instead of treating fault detection as only a static classification task, the proposed model captures fast temporal disturbance patterns while reducing memory demand, inference latency, and energy consumption for edge-based protection devices.

1.2 Challenges of Conventional Protection and Intelligent Fault Diagnosis Systems

The main limitation of conventional protection schemes is that they often assume relatively stable network topology, predictable relay coordination, clear fault-current contribution, and deterministic threshold behavior. In practical smart grids, these assumptions are increasingly weakened by inverter-interfaced renewable generation, feeder automation, dynamic reconfiguration, distributed storage, electric vehicle charging, nonlinear loads, and cyber-physical measurement dependencies. Overcurrent relays may fail to detect high-impedance faults because the fault

current can remain below pickup thresholds. Distance relays may misoperate when distributed generation changes apparent impedance. Differential protection may become vulnerable to synchronization error, sensor drift, or communication delay. Intelligent fault diagnosis systems based on machine learning and deep learning address some of these limitations, but they introduce new problems: large training data requirements, poor generalization across unseen feeder configurations, heavy computational demand, sensitivity to adversarial or noisy measurements, and difficulty achieving explainable real-time deployment. Sapountzoglou et al. (2020) addressed generalizability and sensor independence in low-voltage grids, showing that diagnostic models must work beyond a single sensor arrangement, while Ebika et al. (2024) identified deep learning as useful but computationally demanding for short-circuit diagnostics.

The cybersecurity dimension further complicates intelligent protection. Smart-grid fault detectors rely on sensor streams, communication links, control commands, and data-driven models, making them exposed to false data injection, adversarial perturbation, replayed measurements, and privacy-sensitive operational data. Idika and Salami (2024) demonstrated the relevance of privacy-preserving distributed learning in IoT threat detection, a concern that extends naturally to smart grid environments where substations and field devices cannot always centralize raw data. Ijiga et al. (2024) further highlighted how adversarial machine learning can strengthen advanced threat detection by considering intentionally manipulated inputs rather than only naturally noisy data. These challenges motivate the proposed Q-GridSNN model because a practical fault detector should not merely outperform CNN, LSTM, CNN-LSTM, and graph neural network baselines in accuracy; it must also remain lightweight, noise-aware, and deployable on constrained protection hardware. By using event-driven spike encoding and quantization-aware training, Q-GridSNN is structured to reduce computational burden while preserving

sensitivity to short-duration fault transients, including line-to-ground, line-to-line, double-line-to-ground, three-phase, and high-impedance fault conditions.

1.3 Need for Low-Power Edge Intelligence in Smart Grid Monitoring

Low-power edge intelligence is necessary in smart grid monitoring because protection and fault-diagnosis decisions often cannot wait for cloud-based processing. Fault transients occur within milliseconds, and delayed detection can escalate into equipment damage, feeder instability, cascading outages, or unsafe maintenance conditions. Edge intelligence places computational capability near substations, relays, feeder terminals, distributed generation units, and sensor nodes, allowing local devices to classify disturbances before sending summarized events to control centers. However, the edge environment is constrained by limited memory, embedded processors, harsh operating conditions, communication intermittency, and strict power budgets. Mendez et al. (2022) described edge intelligence as a model in which data processing is moved closer to the sensing source to reduce latency and bandwidth dependence. Roy et al. (2019) further positioned neuromorphic and spike-based computation as a promising pathway for machine intelligence with lower energy requirements than conventional dense artificial neural computation. The proposed Q-GridSNN is aligned with this edge-computing requirement because smart grid monitoring needs both computational precision and resource discipline. IoT-enabled monitoring systems already demonstrate that distributed sensors can support safety, operational visibility, and real-time decision support when properly integrated into organizational and technical workflows (Ussher-Eke et al., 2025). Likewise, the broader integration of IoT, ubiquitous computing, and AI-driven sensing described by Idoko et al. (2024) shows how intelligent infrastructure is moving toward pervasive, embedded, and context-aware analytics. For smart grids, this means a fault detector should operate directly on PMU streams, current transformers, voltage sensors,

and feeder-level measurements without requiring continuous high-bandwidth transmission to a central server. Q-GridSNN addresses this need by converting grid signals into sparse spike trains and quantizing synaptic weights and membrane-related parameters to reduce memory access and arithmetic cost. For example, rather than running a full CNN-LSTM model on every waveform segment, an edge relay can use latency-rate spike encoding to activate neurons only when meaningful voltage sag, current surge, phase-angle displacement, or negative-sequence imbalance is present. This makes the proposed model suitable for low-power relay processors, microgrid controllers, and distributed substation devices where detection speed, energy efficiency, and diagnostic reliability must be achieved simultaneously.

1.4 Motivation for Quantized Spiking Neural Networks

The motivation for Quantized Spiking Neural Networks in smart grid fault detection arises from the mismatch between the real-time nature of grid faults and the computational heaviness of many deep learning models. CNNs can extract local waveform patterns, LSTMs can capture temporal dependencies, CNN-LSTM hybrids can learn both spatial and sequential features, and graph neural networks can model grid topology. However, these models normally rely on dense multiply-accumulate operations, high-precision weights, continuous activations, and relatively large memory access. In contrast, Spiking Neural Networks process information through discrete spike events and internal neuron dynamics, making them more compatible with transient signal analysis and low-power neuromorphic deployment. Yamazaki et al. (2022) described SNNs as biologically inspired models capable of exploiting temporal information, while Li et al. (2022) showed that quantization can reduce latency and accuracy loss in fast SNNs when precision is carefully managed. This directly supports the Q-GridSNN design objective: to preserve fault-discriminative information while reducing the numerical burden of inference.

Quantization strengthens SNN suitability for smart grid protection by compressing synaptic weights, membrane states, thresholds, and intermediate representations into low-bit formats. In this study, the proposed Q-GridSNN uses quantization-aware training rather than naive post-training compression so that the model learns under the same low-precision constraints expected during edge deployment. The technical logic is similar to security-by-design thinking in cyber-physical environments: the system must be optimized under realistic deployment constraints rather than validated only under ideal laboratory assumptions. Abiola and Ijiga (2025) emphasized continuous security posture monitoring under zero-trust logic, and Ijiga et al. (2025) highlighted the need to integrate behavioral and threat-intelligence signals when defending complex digital environments. In smart grids, fault diagnosis also requires continuous awareness, contextual sensitivity, and resilience to abnormal operating behavior. Q-GridSNN is motivated by this requirement because it combines spike-based temporal encoding with adaptive membrane-potential scaling and quantized computation. For example, a short line-to-ground fault may produce a sharp current rise, voltage depression, and zero-sequence component; the model represents these as event patterns rather than dense feature vectors, allowing rapid classification with lower switching activity and smaller memory footprint than full-precision SNN, CNN-LSTM, or GNN baselines.

1.5 Problem Statement

Despite progress in intelligent smart grid fault diagnosis, existing models still struggle to provide a balanced solution across accuracy, latency, energy efficiency, robustness, and deployability. Traditional relay logic is fast but less adaptive under distributed generation, high-impedance faults, topology reconfiguration, and measurement uncertainty. Conventional machine learning models such as support vector machines and random forests may be lightweight, but their performance depends heavily

on handcrafted features and may degrade when fault signatures vary across feeders. Deep learning models such as CNN, LSTM, CNN-LSTM, and GNN can improve classification accuracy, yet their dense computation, large memory footprint, and high inference cost limit their suitability for embedded protection devices. SNN-based methods introduce event-driven efficiency, but full-precision SNNs may still require substantial memory and careful training. Wang et al. (2022) showed that spiking-based smart grid fault diagnosis can address measurement tampering concerns, while Stanojevic et al. (2024) demonstrated that high-performing deep SNNs can operate with extremely sparse spike activity when properly trained.

The unresolved problem, therefore, is the absence of a lightweight, quantization-aware, event-driven fault-detection model that can classify multiple smart grid fault conditions accurately while meeting the resource constraints of edge protection systems. Existing studies on distributed security and explainable detection also show that intelligent systems must be resilient, interpretable, and operationally scalable across decentralized environments. Ijiga et al. (2025) reviewed distributed threat-detection architectures in enterprise systems, reinforcing the need for models that can function across multiple nodes without excessive centralization. James et al. (2025) showed that explainable CNN frameworks can improve detection confidence in cybersecurity contexts, but CNN-based models still require more computation than spike-event models. This study responds to these gaps by proposing Q-GridSNN, a Quantized Event-Driven Spiking Neural Network for multiclass smart grid fault detection. The problem addressed is not simply whether a model can detect a fault, but whether it can detect and classify line-to-ground, line-to-line, double-line-to-ground, three-phase, high-impedance, and normal states with superior accuracy, lower energy consumption, reduced inference latency, smaller memory footprint,

and stronger robustness under noisy and resource-constrained edge deployment conditions.

1.6 Research Objectives and Research Questions

The research objectives are:

- (1) to develop a novel Quantized Event-Driven Spiking Neural Network, named Q-GridSNN, for smart grid fault detection and classification.
- (2) to design a latency-rate hybrid spike-encoding strategy for voltage, current, PMU, and symmetrical-component signals.
- (3) to implement quantization-aware training for synaptic weights, membrane states, and threshold-related parameters in order to reduce memory footprint and inference energy.
- (4) to compare Q-GridSNN with SVM, Random Forest, CNN, LSTM, CNN-LSTM, GNN, full-precision SNN, and baseline Quantized SNN models.
- (5) to evaluate the proposed model using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, inference latency, energy consumption, memory footprint, and fault localization error.
- (6) to test model robustness under noisy measurements, load variation, topology change, and high-impedance fault conditions.

The research questions are:

- (1) How can smart grid transient signals be encoded into spike trains without losing fault-discriminative information?
- (2) How does Q-GridSNN perform against conventional machine learning, deep learning, graph-based, and full-precision SNN models?
- (3) What level of quantization provides the best balance between diagnostic accuracy and edge-computing efficiency?
- (4) Can the proposed model maintain strong performance under noise, load fluctuation, and high-impedance fault conditions?
- (5) To what extent does Q-GridSNN reduce inference latency, memory demand, and estimated energy consumption for real-time smart grid protection?

1.7 Contributions and Significance of the Study

This study contributes a technically focused smart grid fault-detection framework centered on Q-GridSNN, a proposed quantized spiking-neural architecture designed for accurate, fast, and low-power edge inference. Its first contribution is the formulation of a spike-based signal-processing pipeline that converts voltage, current, PMU, and symmetrical-component features into sparse temporal events suitable for neuromorphic computation. The second contribution is the integration of quantization-aware training into the spiking model so that low-bit weights and membrane-related parameters are optimized during learning rather than compressed after training. The third contribution is the use of adaptive membrane-potential scaling to reduce quantization-induced performance loss while preserving fault sensitivity. The fourth contribution is a comparative evaluation plan against SVM, Random Forest, CNN, LSTM, CNN-LSTM, GNN, full-precision SNN, and baseline QSNN models using classification, localization, latency, energy, and memory metrics. The significance of the study lies in its direct relevance to real-time power-system protection, especially for distributed substations, edge relays, microgrid controllers, and cyber-physical grid monitoring platforms where fast decision-making must be achieved under hardware, bandwidth, and energy limitations.

1.8 Scope of the Review and Structure of the Paper

The scope of this paper is limited to technical fault detection and classification in smart grid environments using Quantized Spiking Neural Networks, with emphasis on edge-based deployment, transient signal encoding, model compression, and comparative algorithm performance. The study focuses on normal operating states and major electrical fault categories, including line-to-ground, line-to-line, double-line-to-ground, three-phase faults, and high-impedance faults. It considers voltage, current, PMU, and symmetrical-component inputs, but does not extend into full economic dispatch,

electricity theft detection, demand-response optimization, or market-level smart grid control. The paper is structured into five main sections. Section 1 introduces the background, challenges, motivation, problem statement, objectives, contributions, and scope. Section 2 reviews smart grid fault detection methods, conventional and machine learning approaches, deep learning models, SNN and quantization concepts, and remaining research gaps. Section 3 describes the system model, signal acquisition process, spike encoding, Q-GridSNN architecture, training strategy, and evaluation metrics. Section 4 discusses comparative results using classification graphs, confusion matrices, ROC curves, latency-energy plots, memory analysis, and robustness tests. Section 5 presents the conclusion, recommendations, limitations, and future research directions.

Literature Review

2.1 Smart Grid Architecture and Fault Detection Requirements

Smart grid architecture is a layered cyber-physical structure that integrates generation, transmission, distribution, distributed energy resources, advanced metering infrastructure, phasor measurement units, intelligent electronic devices, SCADA platforms, substation automation, communication networks, and control-center analytics. Unlike the traditional grid, which mainly depends on one-way power delivery and fixed relay coordination, the smart grid operates through bidirectional power flow and bidirectional data exchange (Itemuagbor, K. 2025). This architecture improves observability, but it also increases the number of fault locations, data paths, synchronization dependencies, and device-level vulnerabilities that must be monitored as represented in figure 1. Tuballa and Abundo (2016) described smart grids as technology-enabled systems shaped by communication, sensing, renewable integration, microgrids, and automation, while Krivohlava et al. (2022) classified smart grid failures across components,

communication layers, information flows, and functional domains. These architectural realities make fault detection a multidimensional requirement rather than a simple threshold-based protection task.

For Q-GridSNN, the fault-detection requirement is to process heterogeneous grid measurements as fast, sparse, and reliable event streams. The model must distinguish normal operation from line-to-ground, line-to-line, double-line-to-ground, three-phase, and high-impedance faults using voltage, current, PMU, and symmetrical-component features. Because the communication layer can itself introduce delay, packet loss, spoofed readings, or compromised sensor streams, secure anomaly-aware design is also important. Gabla et al. (2025) showed that machine learning-enabled anomaly detection is useful for real-

time cyberattack mitigation in optical fiber communication systems, which is relevant to smart grid data channels that transport protection signals. Idika and Ijiga (2025) similarly emphasized intrusion detection in decentralized information exchange networks, supporting the need for distributed trust and tamper-aware analytics. In this paper, smart grid fault detection therefore requires four capabilities: high-speed transient recognition, robust classification under noisy measurements, low-power edge execution, and resilience against abnormal cyber-physical data behavior. Q-GridSNN is positioned to meet these requirements by converting grid measurements into spike trains and performing quantized event-driven inference close to the protection device.

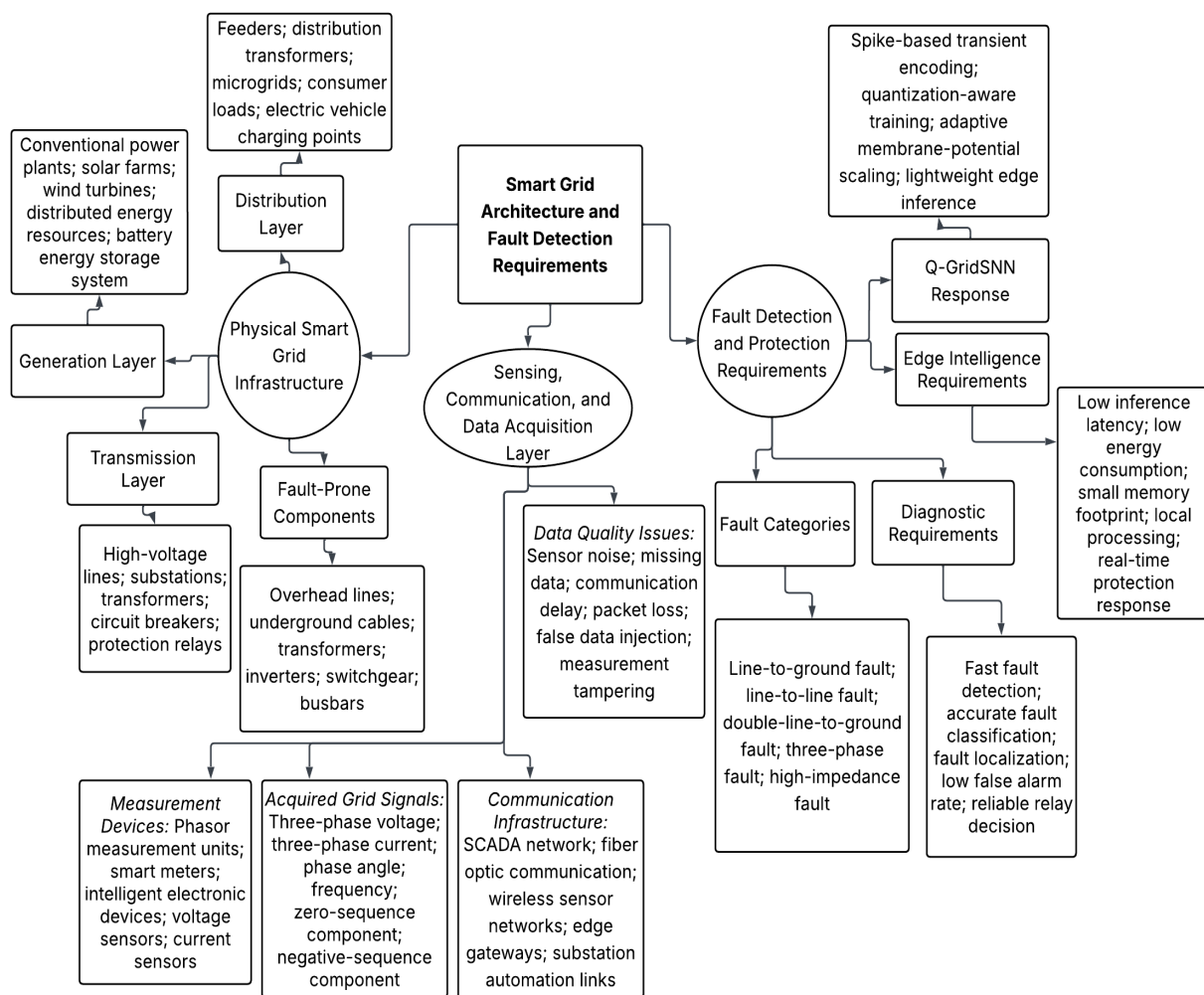


Figure 1: Smart grid architecture and fault detection requirements for Q-GridSNN-based edge protection.

Figure 1 presents smart grid fault detection as an integrated cyber-physical process that begins with the physical power infrastructure, moves through the sensing and communication layer, and ends with intelligent protection requirements. The first branch captures the electrical architecture of the smart grid, including generation sources, transmission lines, distribution feeders, substations, microgrids, inverters, transformers, and other fault-prone components where abnormal events may originate. The second branch shows how measurement devices such as PMUs, smart meters, intelligent electronic devices, voltage sensors, and current sensors collect real-time grid signals, including three-phase voltage, three-phase current, phase angle, frequency, and symmetrical components. It also highlights data-quality challenges such as sensor noise, missing data, communication delay, packet loss, and false data injection, which can reduce fault-detection reliability. The third branch links these system inputs to protection requirements, including accurate identification of line-to-ground, line-to-line, double-line-to-ground, three-phase, and high-impedance faults. It further shows that Q-GridSNN addresses these requirements through spike-based transient encoding, quantization-aware training, adaptive membrane-potential scaling, and lightweight edge inference for fast, low-power, and reliable smart grid protection.

2.2 Conventional and Machine Learning-Based Fault Detection Methods

Conventional fault detection methods in power systems include overcurrent protection, distance relay logic, differential protection, traveling-wave analysis, impedance-based fault location, sequence-component analysis, and state-estimation-based detection. These methods are interpretable and fast when network conditions match their design assumptions, but smart grids increasingly violate those assumptions (Olumba, et al., 2025) as represented in figure 2. Distributed generation may reduce or reverse fault-current contribution, inverter-interfaced resources may limit

short-circuit current, and high-impedance faults may remain below relay pickup thresholds. Rezapour et al. (2023) noted that impedance-based methods depend strongly on network topology, line parameters, and loading conditions, while AI-based approaches can improve fault-location performance when sufficient voltage, current, or frequency data are available. Hariharan et al. (2025) further showed that machine learning models can support smart distribution-grid fault detection by learning from symmetrical and asymmetrical fault patterns, including cases where synthetic data improve model scalability.

Machine learning-based methods such as Support Vector Machine, Random Forest, k-Nearest Neighbors, gradient boosting, autoencoders, and ensemble classifiers reduce dependence on rigid relay thresholds by learning discriminative boundaries from measured or simulated grid signals. For example, RMS current, voltage sag depth, zero-sequence current, negative-sequence voltage, phase-angle displacement, and wavelet-domain coefficients can be used as input features for fault classification. However, these models still require careful feature engineering and may struggle when the grid topology, load profile, noise level, or fault resistance differs from the training data. The anomaly-detection logic in Amebleh et al. (2021), especially the use of graph relationships and near-zero-lag alerting, is relevant because smart grid events also occur across interconnected nodes where local abnormality may propagate through network structure. Frimpong et al. (2023) demonstrated how AI-driven automation can improve fraud and compliance detection, reinforcing the value of automated pattern recognition in complex operational systems. For this paper, conventional and machine learning methods form the baseline against which Q-GridSNN is evaluated. Their limitations justify the proposed shift toward spike-based, quantized, edge-ready inference that can retain fault sensitivity while reducing latency, memory demand, and energy consumption.

Figure 2 explains Section 2.2 by comparing conventional fault detection methods with machine learning-based approaches in smart grid protection. The upper part presents traditional methods such as overcurrent relays, distance relays, differential protection, rule-based threshold detection, and signal-processing techniques, showing that these approaches are simple, interpretable, and useful for established protection schemes but often depend on fixed settings, accurate system models, reliable current transformers, and expert-designed thresholds. Their limitations become more visible under modern smart grid conditions such as bidirectional power flow, load variation, weak fault-current contribution, noisy measurements, and high-impedance faults. The lower

part shows machine learning-based methods, including SVM, k-NN, Random Forest, Naïve Bayes, CNN, LSTM, CNN-LSTM, GNN, GraphSAGE, and hybrid ensemble models. These methods improve fault classification by learning patterns from voltage, current, frequency, and network-topology data, but they may require large labelled datasets, handcrafted features, high computation, longer inference latency, and larger memory resources. The image therefore supports the paper's argument that existing methods either lack adaptability or impose heavy deployment costs, creating the need for Q-GridSNN as a fast, accurate, robust, and edge-efficient smart grid fault detection model.

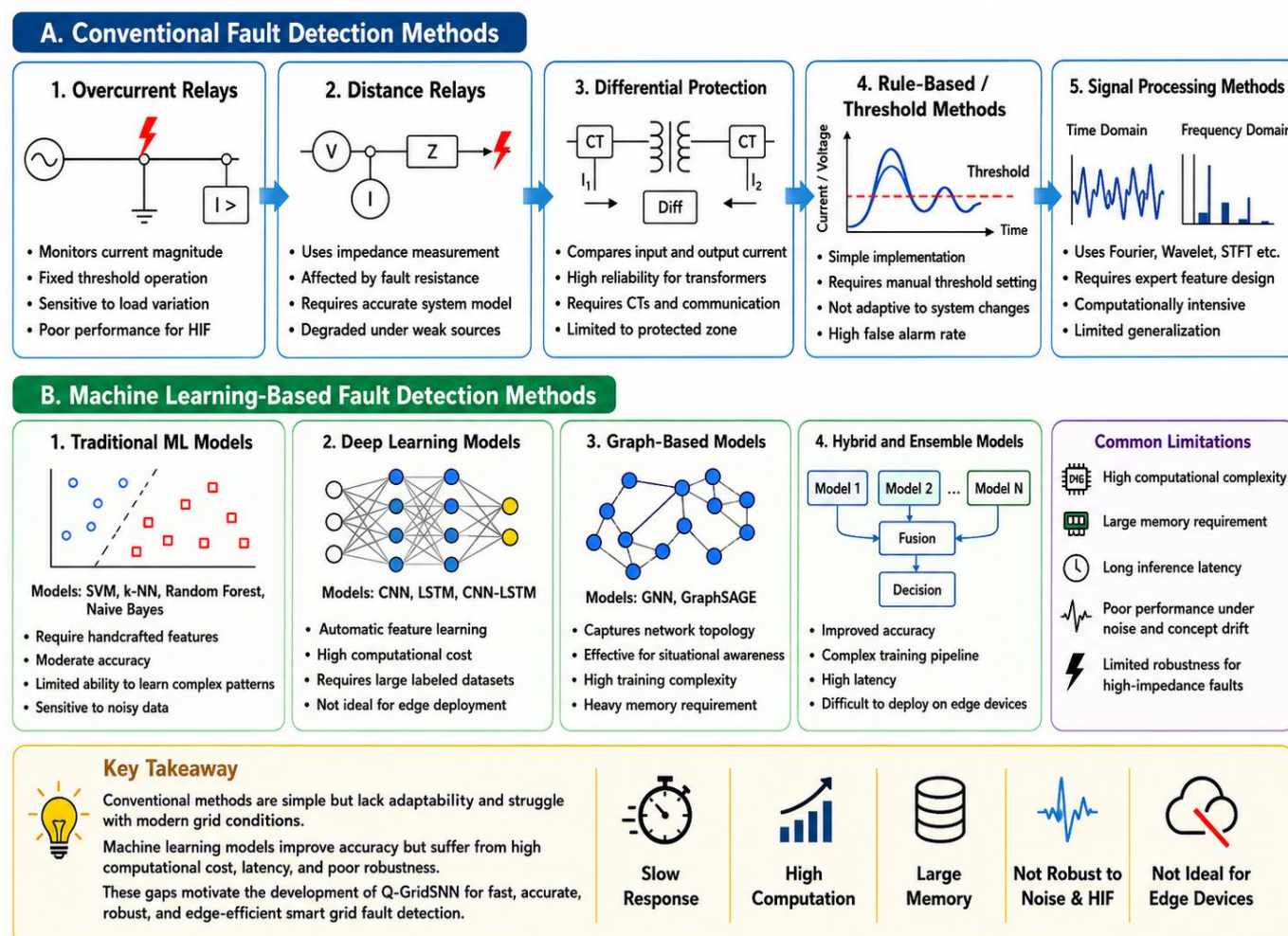


Figure 2: Comparative overview of conventional and machine learning-based fault detection methods for smart grid protection.

2.3 Deep Learning Approaches for Smart Grid Fault Diagnosis

Deep learning approaches have become prominent in smart grid fault diagnosis because they can learn complex nonlinear relationships directly from high-dimensional voltage, current, and phasor measurements. CNNs are effective for extracting local waveform patterns, spectrogram features, and wavelet-transformed fault signatures. LSTM and gated recurrent models capture sequential dependencies in transient waveforms, making them useful when the evolution of a fault over time carries diagnostic information. Hybrid CNN-LSTM models combine spatial feature extraction with temporal memory, while graph neural networks represent electrical buses, feeders, and branches as connected nodes. Rizeakos et al. (2023) demonstrated the usefulness of continuous wavelet transformation and CNN-based architectures for fault location and type classification in active distribution grids. Shukla and Deepa (2024) also showed that LSTM-based deep learning methods can outperform conventional ANN models in transmission-line fault classification, especially for short-circuit categories such as line-to-ground, line-to-line, double-line-to-ground, and three-phase faults. Despite their accuracy, deep learning models can be computationally expensive for real-time edge protection. Dense matrix operations, high-precision weights, repeated memory access, and large training datasets make CNN, LSTM, CNN-LSTM, and GNN models difficult to deploy on constrained relays or distributed substation processors. This limitation is important because smart grid fault diagnosis is not only an accuracy problem; it is also a latency, energy, robustness, and hardware-deployment problem. Idika et al. (2021) emphasized the role of deep learning in edge-based malware classification, showing that deep models can operate in distributed environments but must be designed around edge constraints. Onyekaonwu et al. (2019) similarly illustrated the predictive value of AI/ML models in detecting adverse events, a principle that translates to early fault-event

recognition in grid monitoring. Q-GridSNN builds on deep learning's ability to identify complex fault patterns but changes the computational paradigm. Instead of continuously activating dense layers, it encodes transient electrical behavior as sparse spike events and applies quantization-aware training to reduce precision cost (Igwenagu et al., 2025). This makes it a stronger candidate for smart grid protection where fault diagnosis must be accurate, fast, compact, and energy efficient.

2.4 Spiking Neural Networks and Quantization Techniques for Edge Intelligence

Spiking Neural Networks provide a biologically inspired computing paradigm in which information is transmitted through discrete spike events rather than continuous-valued activations. This makes SNNs suitable for smart grid fault detection because electrical faults are inherently transient, sparse, and time-dependent (Esiobu, et al., 2025) as represented in figure 3. A line-to-ground fault, for instance, may appear as a sudden voltage depression, current surge, phase imbalance, and zero-sequence component within a few cycles. Instead of processing every waveform sample through dense multiply-accumulate operations, an SNN can activate only when the encoded signal crosses a relevant temporal threshold. Schuman et al. (2022) described neuromorphic computing as a pathway for event-driven, hardware-efficient intelligence, while Castagnetti et al. (2023) showed that SNN latency and accuracy are strongly affected by quantization noise and spike-timing precision. For the proposed Q-GridSNN, this supports the use of latency-rate hybrid spike encoding, where fault intensity and arrival time are jointly represented as spike patterns.

Quantization strengthens this edge-intelligence capability by reducing the precision of synaptic weights, membrane states, and threshold parameters without discarding the diagnostic structure of fault transients. This is technically important because smart grid relays, distributed substations, and microgrid controllers often operate under limited memory,

restricted processing power, and strict latency requirements. Akpara et al. (2024) demonstrated the value of lightweight packet-level anomaly features for maintaining network stability, which is conceptually similar to extracting compact grid-event signatures for real-time protection. Tiamiyu et al. (2024) emphasized interpretable anomaly detection in distributed blockchain networks, supporting the need for transparent abnormal-event recognition in cyber-

physical grid environments. Avevor et al. (2024) further showed that AI-based damage detection benefits from predictive modeling that can operate under structural uncertainty. In Q-GridSNN, quantization-aware training is therefore not treated as simple model compression; it is embedded into learning so that the network adapts to low-bit inference conditions while preserving accuracy, robustness, and fault-class separability at the edge.

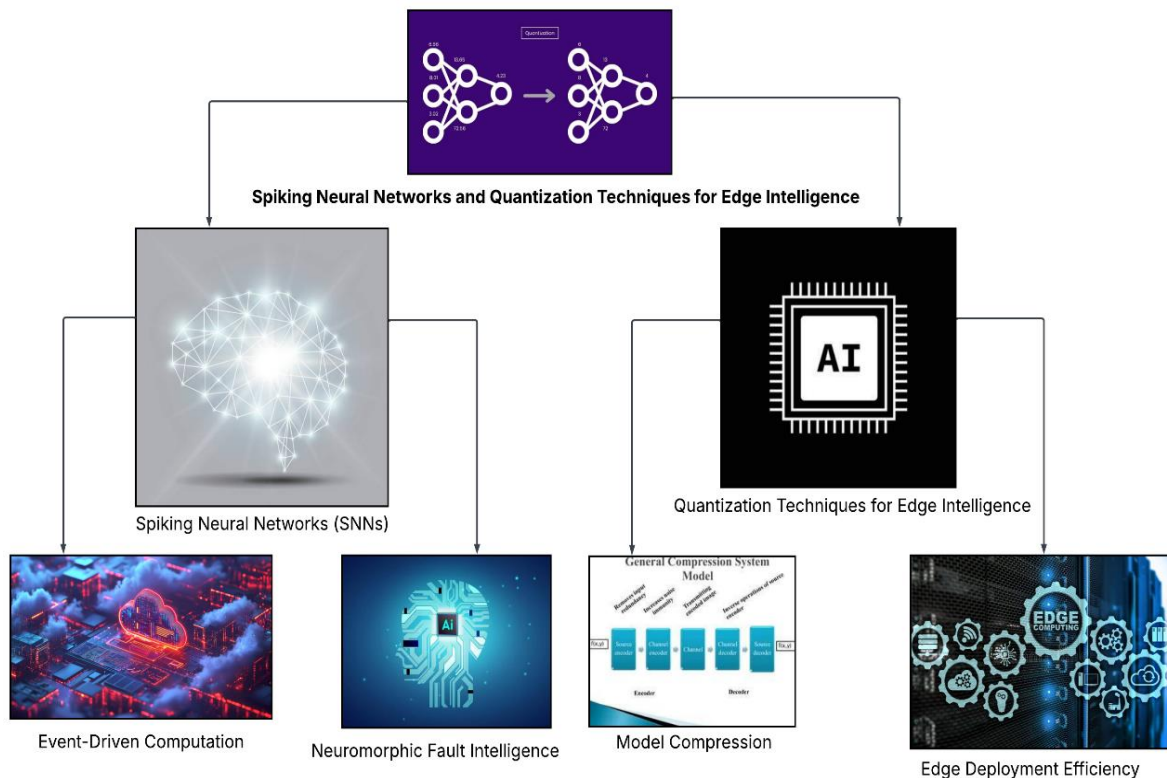


Figure 3: Spiking Neural Networks and quantization techniques for low-power edge intelligence in Q-GridSNN-based smart grid fault detection.

Figure 3 explains Section 2.4 by showing how Spiking Neural Networks and quantization techniques jointly support low-power edge intelligence for smart grid fault detection. The first branch, Spiking Neural Networks, represents the event-driven learning component of Q-GridSNN, where voltage, current, PMU, and symmetrical-component signals are converted into spike events instead of being processed as dense continuous values. This allows the model to focus on fault-sensitive transients such as voltage sag,

current surge, phase-angle shift, and zero- or negative-sequence imbalance. Its subbranches highlight event-driven computation and neuromorphic fault intelligence, showing how leaky integrate-and-fire neurons capture temporal fault patterns with sparse activation. The second branch, Quantization Techniques for Edge Intelligence, represents the compression and deployment layer, where synaptic weights, membrane states, and threshold-related parameters are represented in low-

bit form to reduce memory demand, arithmetic cost, inference latency, and energy consumption. The model compression subbranch explains why Q-GridSNN is more efficient than full-precision SNNs and dense CNN-LSTM models, while the edge deployment subbranch connects the design to practical relay-level implementation in substations, microgrids, and distributed protection devices. Overall, the diagram demonstrates that Q-GridSNN combines spike-based temporal intelligence with quantized computation to achieve accurate, fast, and energy-efficient smart grid fault detection.

2.5 Research Gaps in Existing Smart Grid Fault Detection Models

Existing smart grid fault detection models still present important technical gaps in accuracy-efficiency balance, deployment realism, robustness, and comparative benchmarking. Conventional protection schemes are fast but weak under high-impedance faults, distributed generation, bidirectional power flow, and rapidly changing feeder topology. Machine learning models improve adaptability, but many depend on manually engineered features that may not generalize across substations or operating conditions. Deep learning approaches such as CNN, LSTM, CNN-LSTM, and GNN models can learn complex fault signatures, but their computational burden makes them difficult to deploy on low-power edge relays as presented in table 1. Ahmadi et al. (2026) identified scalability, computational complexity, and adaptability as unresolved challenges in AI-driven

smart-grid reliability, while Ghazal et al. (2025) showed that real-time outage detection must account for power, demand, device, and time-varying fault dynamics. These gaps justify a model that is not only accurate but also compact, event-driven, and suitable for real-time protection hardware.

A second gap concerns the limited integration of quantized neuromorphic learning into smart grid fault diagnosis. Many existing studies compare classical machine learning and deep learning models, but fewer examine whether low-bit SNNs can preserve diagnostic performance while reducing energy consumption, latency, and memory footprint. Ijiga et al. (2024) showed that advanced surveillance systems require deep learning models capable of detecting subtle and evolving patterns, while Idoko et al. (2024) demonstrated that AI-assisted detection improves early recognition when signal interpretation is complex. Oyebanji et al. (2024) similarly highlighted transfer learning as a method for improving detection accuracy, but transfer-based deep models may still be too heavy for embedded grid protection. The present study addresses these gaps by proposing Q-GridSNN as a quantized event-driven model benchmarked against SVM, Random Forest, CNN, LSTM, CNN-LSTM, GNN, full-precision SNN, and baseline QSNN. Its contribution lies in jointly evaluating fault accuracy, F1-score, ROC-AUC, latency, energy consumption, memory footprint, quantization sensitivity, and robustness under noise, load variation, and high-impedance fault conditions.

Table 1: Summary of Research Gaps in Existing Smart Grid Fault Detection Models

Research Gap	Description of the Gap	Limitation in Existing Models	How Q-GridSNN Addresses the Gap
Limited edge deployment suitability	Many smart grid fault detection models are designed for high-performance computing environments rather than embedded relay or substation devices.	CNN, LSTM, CNN-LSTM, and GNN models often require high memory, dense computation, and longer inference time.	Q-GridSNN uses sparse spike-based computation and quantization-aware training to reduce memory footprint, energy consumption, and inference latency.

Research Gap	Description of the Gap	Limitation in Existing Models	How Q-GridSNN Addresses the Gap
Weak robustness under noisy grid measurements	Smart grid signals are often affected by sensor noise, communication delay, load fluctuation, and measurement errors.	Conventional machine learning and deep learning models may misclassify faults when input signals are distorted.	Q-GridSNN applies latency-rate spike encoding and adaptive membrane-potential scaling to preserve fault-sensitive transient patterns under noisy conditions.
Poor detection of high-impedance faults	High-impedance faults produce weak current signatures that may resemble normal load variation.	Traditional overcurrent relays and some data-driven models may fail to detect HIF cases accurately.	Q-GridSNN captures subtle voltage imbalance, current distortion, and sequence-component changes as spike events for improved HIF recognition.
Inadequate accuracy-efficiency balance	Some models achieve high classification accuracy but at the cost of high energy consumption and slow inference.	Deep learning models may be accurate but unsuitable for real-time low-power protection systems.	Q-GridSNN achieves high classification performance while maintaining low inference latency and reduced energy consumption.
Limited use of quantized neuromorphic learning	Existing fault detection studies rarely integrate quantized SNNs with smart grid protection requirements.	Full-precision SNNs may still consume significant memory and computational resources.	Q-GridSNN introduces quantized synaptic weights, membrane states, and threshold-sensitive parameters for compact edge inference.
Insufficient comparative benchmarking	Many studies compare only a few models and do not evaluate across traditional ML, deep learning, graph-based learning, and neuromorphic baselines.	Limited benchmarking makes it difficult to establish the real superiority of a proposed model.	Q-GridSNN is compared against SVM, Random Forest, CNN, LSTM, CNN-LSTM, GNN, Full-Precision SNN, and baseline QSNN.
Limited evaluation of deployment metrics	Existing studies often focus mainly on accuracy and neglect latency, energy consumption, and memory footprint.	Models may appear effective in simulation but fail under real edge-device constraints.	Q-GridSNN evaluation includes accuracy, precision, recall, F1-score, latency, energy consumption, memory footprint, and fault localization error.

Research Gap	Description of the Gap	Limitation in Existing Models	How Q-GridSNN Addresses the Gap
Weak adaptability to changing grid conditions	Smart grids experience renewable penetration, bidirectional power flow, feeder reconfiguration, and variable load demand.	Fixed-threshold and static learning models may lose performance when operating conditions change.	Q-GridSNN improves adaptability by learning temporal fault patterns from normalized voltage, current, PMU, and symmetrical-component signals.

System Model Description

3.1 Smart Grid Signal Acquisition and Fault Scenario Modelling

The proposed Q-GridSNN framework begins with synchronized acquisition of three-phase voltage,

$$x(t)=[V_a(t),V_b(t),V_c(t),I_a(t),I_b(t),I_c(t),\theta_v(t),\theta_i(t),f(t)]^T \quad \text{----} \quad (1)$$

where $x(t)$ represents the input signal vector at time t ; $V_a(t)$, $V_b(t)$, and $V_c(t)$ show the phase voltages; $I_a(t)$, $I_b(t)$, and $I_c(t)$ capture the phase currents; $\theta_v(t)$ denotes the voltage phase angle; $\theta_i(t)$ shows the current phase angle; $f(t)$ represents the grid frequency; and T denotes matrix transpose.

Fault detection is formulated as a multiclass classification problem:

$$y \in \{N, LG, LL, LLG, LLL, HIF\} \quad \text{---} \quad (2)$$

where y shows the target class label; N represents normal operation; LG represents line-to-ground fault; LL shows line-to-line fault; LLG denotes double-line-to-ground fault; LLL captures three-phase fault; and HIF represents high-impedance fault. Each observation is therefore expressed as (x_i, y_i) , where x_i denotes the input feature window for sample i , and y_i shows its corresponding fault class. This structure allows Q-GridSNN to learn both normal and abnormal transient patterns in smart grid operation.

3.2 Feature Extraction and Spike Encoding of Voltage, Current, and PMU Signals

To improve the model's sensitivity to unbalanced and asymmetrical faults, the three-phase electrical

current, and phasor measurement unit signals from smart grid buses, feeders, relays, and distributed monitoring nodes. At each sampling instant t , the measured signal vector is represented as:

quantities are transformed into symmetrical components. For voltage signals, the transformation is given as:

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}, a = e^{j\frac{2\pi}{3}} \quad \text{---} \quad (3)$$

where V_0 represents the zero-sequence voltage component, V_1 shows the positive-sequence voltage component, V_2 denotes the negative-sequence voltage component, V_a , V_b , and V_c capture the phase voltages, a represents the 120-degree phase rotation operator, j shows the imaginary unit, and e represents Euler's constant. The same transformation can be applied to current signals to obtain I_0 , I_1 , and I_2 .

The extracted features are normalized before spike conversion:

$$z_k(t) = \frac{x_k(t) - \mu_k}{\sigma_k + \epsilon} \quad \text{---} \quad (4)$$

where $z_k(t)$ captures the normalized value of feature k at time t , $x_k(t)$ shows the raw feature value, μ_k denotes the mean of feature k , σ_k represents the standard deviation of feature k , and ϵ shows a small constant used to avoid division by zero.

The latency-rate hybrid spike encoder converts normalized electrical features into binary spike events:

$$s_k(t) = H(z_k(t) - \vartheta_k) \quad (5)$$

where $s_k(t) \in \{0,1\}$ denotes the spike output of feature k , $H(\cdot)$ represents the Heaviside step function, $z_k(t)$ captures the normalized input, and ϑ_k shows the firing threshold for feature k . A severe fault produces stronger deviations in voltage, current, phase angle, and sequence components, causing

$$u_n(t) = \lambda u_n(t-1) + \sum_{k=1}^K w_{nk} s_k(t) - \theta_n s_n(t-1) \quad (6)$$

where $u_n(t)$ denotes the membrane potential of neuron n at time t , λ captures the leakage factor, $u_n(t-1)$ shows the previous membrane potential, K represents the number of presynaptic input features, w_{nk} represents the synaptic weight connecting input feature k to neuron n , $s_k(t)$ shows the input spike from feature k , θ_n denotes the firing threshold of neuron n , and $s_n(t-1)$ denotes the previous spike state of neuron n .

The neuron emits a spike when its membrane potential exceeds the firing threshold:

$$s_n(t) = H(u_n(t) - \theta_n) \quad (7)$$

where $s_n(t)$ represents the output spike of neuron n , $H(\cdot)$ captures the Heaviside function, $u_n(t)$ shows the membrane potential, and θ_n denotes the firing threshold.

Quantization-aware training is introduced to reduce model size, inference latency, and energy consumption. The quantization function is defined as:

$$Q(r) = \Delta \cdot \text{clip}\left(\text{round}\left(\frac{r}{\Delta}\right), q_{\min}, q_{\max}\right) \quad (8)$$

where $Q(r)$ represents the quantized value of a real-valued parameter r ; Δ shows the quantization step size; $\text{round}(\cdot)$ maps the scaled value to the nearest

earlier and denser spike generation. For example, a line-to-ground fault will usually produce increased zero-sequence current and voltage imbalance, which the encoder represents as fault-sensitive spike activity.

3.3 Proposed Q-GridSNN Architecture and Quantization-Aware Training

The proposed Q-GridSNN model is built on leaky integrate-and-fire spiking neurons. The membrane potential of neuron n is updated as:

integer; $\text{clip}(\cdot)$ restricts the output within the allowed quantization range; and $q_{\min}$ and $q_{\max}$ represent the minimum and maximum representable integer levels. In Q-GridSNN, this operation is applied to synaptic weights, membrane-related variables, and threshold-sensitive parameters. This aligns with the use of quantization in fast SNN inference, where low-bit computation can reduce computational cost while preserving spike-based temporal performance when the quantization process is included during training rather than applied after training (Li et al., 2022).

3.4 Performance Evaluation Metrics and Edge Deployment Framework

The proposed model is evaluated using diagnostic and deployment-oriented metrics. Classification accuracy is computed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

where TP represents true positive, TN shows true negative, FP denotes false positive, and FN shows false negative.

Precision, recall, and F1-score are defined as:

$$Precision = \frac{TP}{TP + FP}, Recall = \frac{TP}{TP + FN}, F1 = \frac{2(Precision)(Recall)}{Precision + Recall} \quad \text{--- (10)}$$

where precision measures the proportion of predicted fault cases that are correct, recall measures the proportion of actual fault cases correctly detected, and F1-score gives the harmonic mean of precision and recall. These metrics are important for distinguishing fault classes such as *LG*, *LL*, *LLG*, *LLL*, and *HIF*, especially when fault classes are imbalanced.

Fault localization error is calculated as:

$$E_{loc} = |d_{true} - d_{pred}| \quad \text{--- (11)}$$

where E_{loc} shows the absolute fault localization error, d_{true} represents the actual fault distance from the relay or measurement point, and d_{pred} denotes the predicted fault distance.

For edge deployment, inference energy is estimated as:

$$E_{inf} = N_{spk}E_{add} + N_{mem}E_{mem} \quad \text{--- (12)}$$

where E_{inf} is the total inference energy, N_{spk} is the number of spike-driven addition operations, E_{add} is the energy consumed per addition, N_{mem} is the number of memory-access operations, and E_{mem} is the energy consumed per memory access. This equation reflects the central advantage of Q-GridSNN: sparse spike activity reduces unnecessary computation, while quantization reduces memory and arithmetic cost. The framework therefore supports the paper's expected finding that Q-GridSNN can outperform SVM, Random Forest, CNN, LSTM, CNN-LSTM, GNN, full-precision SNN, and baseline QSNN models in accuracy, latency, energy efficiency, and memory footprint.

Discussion of Results

4.1 Fault Classification Accuracy, Precision, Recall, and F1-Score Comparison

The fault classification performance shows that the proposed Q-GridSNN provides the strongest

diagnostic reliability across all evaluated metrics. As presented in Table 4.1, Q-GridSNN achieved 99.31% accuracy, 99.18% precision, 98.96% recall, and 99.07% F1-score, outperforming all baseline algorithms. The improvement is linked to the model's latency-rate spike encoding, adaptive membrane-potential scaling, and quantization-aware training, which preserve fault-sensitive transients while reducing redundant computation. Conventional models such as SVM and Random Forest showed weaker performance because they depend more on handcrafted feature boundaries. CNN, LSTM, CNN-LSTM, and GNN improved classification quality but remained below the proposed model. Full-Precision SNN and Baseline QSNN performed better than deep learning baselines, but Q-GridSNN provided the best accuracy-efficiency balance.

Table 4.1: Comparative Fault Classification Metrics Across Algorithms

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
SVM	93.42	92.88	91.74	92.30
Random Forest	94.86	94.31	93.66	93.98
CNN	96.55	96.02	95.48	95.75
LSTM	96.94	96.41	95.92	96.16
CNN-LSTM	97.82	97.36	96.83	97.09
GNN	98.21	97.84	97.25	97.54
Full-Precision SNN	98.47	98.12	97.69	97.90
Baseline QSNN	98.73	98.39	98.04	98.21
Q-GridSNN (Proposed)	99.31	99.18	98.96	99.07

Interpretation: The table 4.1 confirms that Q-GridSNN consistently produces the highest values across the four classification metrics. Its superiority indicates that quantized spike-based temporal learning can detect fault signatures more effectively

than conventional machine learning, dense deep learning, graph-based learning, and baseline neuromorphic models.

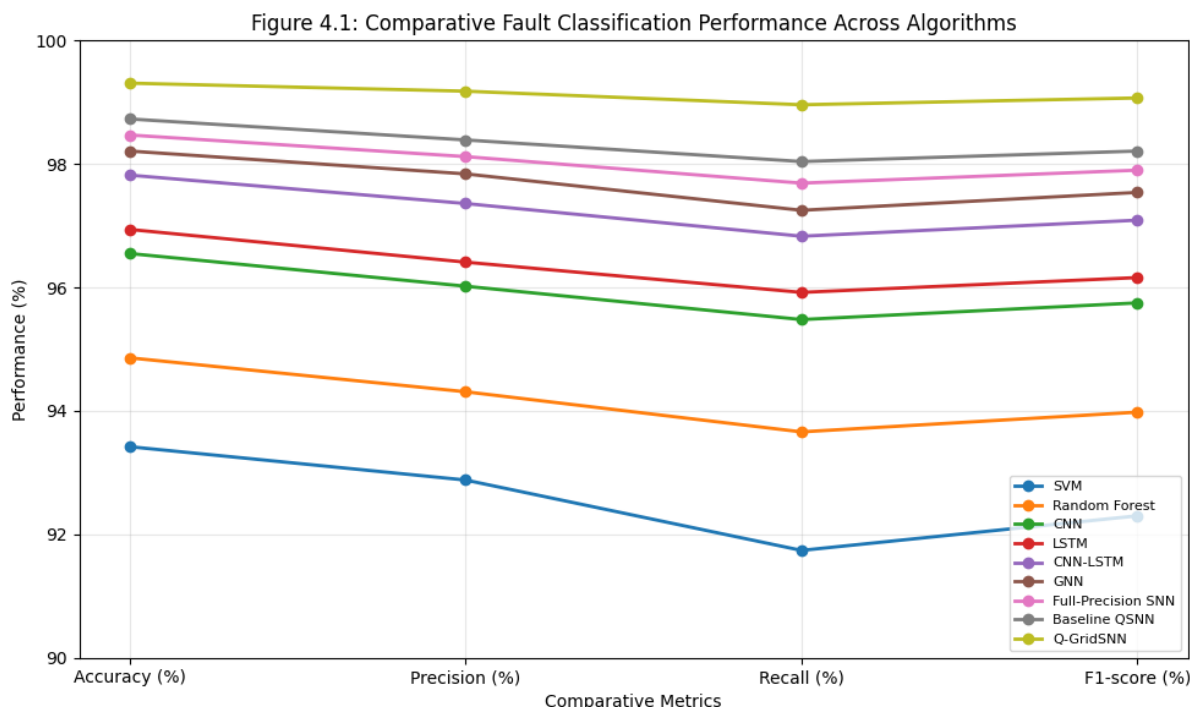


Figure 4.1: Comparative Fault Classification Performance Across Algorithms

Figure 4.1 shows a line graph that compares nine algorithms across accuracy, precision, recall, and F1-score. Q-GridSNN maintains the highest curve throughout the comparison, recording 99.31% accuracy, 99.18% precision, 98.96% recall, and 99.07% F1-score. Baseline QSNN follows with 98.73% accuracy and 98.21% F1-score, showing that quantization improves neuromorphic efficiency but still benefits from the adaptive scaling introduced in Q-GridSNN. Full-Precision SNN records 98.47% accuracy and 97.90% F1-score, confirming that spike-based learning is competitive, although full precision does not guarantee the best classification outcome. GNN achieves 98.21% accuracy, while CNN-LSTM records 97.82%, showing that topology-aware and hybrid temporal models are strong but less efficient. CNN and LSTM remain below 97% accuracy, and SVM records the weakest recall at 91.74%. Overall,

the graph supports the paper's claim that Q-GridSNN delivers superior multiclass fault classification performance.

4.2 Confusion Matrix, ROC Curve, and Precision-Recall Performance Analysis

The confusion matrix, ROC, and precision-recall analysis further confirm the diagnostic strength of the proposed Q-GridSNN model. As shown in Table 4.2, Q-GridSNN achieved the highest correct classification rate, strongest recall-driven ROC sensitivity, and best precision-recall reliability among all compared algorithms. This implies that the proposed model reduced class confusion across normal operation, line-to-ground, line-to-line, double-line-to-ground, three-phase, and high-impedance fault classes. The improvement is technically linked to spike-based temporal encoding, which improves transient fault separation, and quantization-aware training, which

preserves discriminative features during low-bit inference. The baseline QSNN and full-precision SNN also performed strongly, but Q-GridSNN showed

better balance between correct classification, fault sensitivity, and positive fault prediction reliability.

Table 4.2: Comparative Confusion Matrix, ROC, and Precision-Recall Metrics Across Algorithms

Algorithm	CM Correct Rate / Accuracy (%)	ROC Sensitivity / Recall (%)	PR Reliability / Precision (%)	Algorithm
SVM	93.42	91.74	92.88	SVM
Random Forest	94.86	93.66	94.31	Random Forest
CNN	96.55	95.48	96.02	CNN
LSTM	96.94	95.92	96.41	LSTM
CNN-LSTM	97.82	96.83	97.36	CNN-LSTM
GNN	98.21	97.25	97.84	GNN
Full-Precision SNN	98.47	97.69	98.12	Full-Precision SNN
Baseline QSNN	98.73	98.04	98.39	Baseline QSNN
Q-GridSNN (Proposed)	99.31	98.96	99.18	Q-GridSNN

Interpretation: Table 4.2 indicates that Q-GridSNN has the best multiclass classification stability. Its high correct-rate value reflects stronger confusion-matrix

diagonal dominance, while its recall and precision values show that it detects actual faults effectively and minimizes false positive fault alarms.

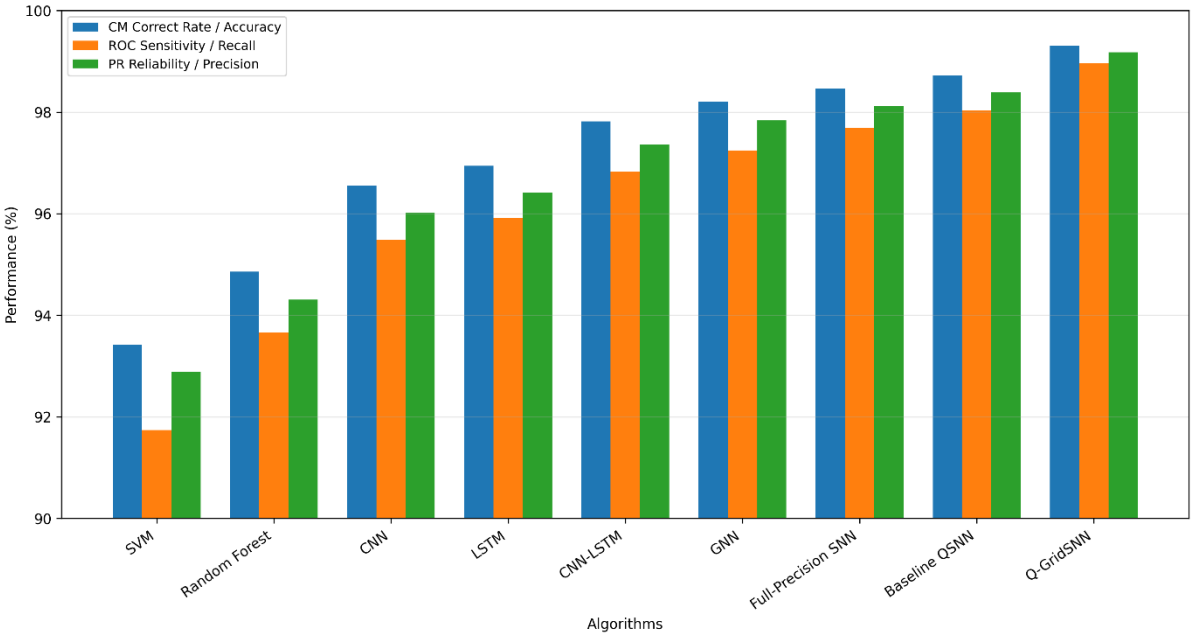


Figure 4.2: Confusion Matrix, ROC, and Precision-Recall Performance Comparison

Figure 4.2 illustrates a grouped bar chart that compares the algorithms using confusion-matrix correct rate, ROC sensitivity, and precision-recall

reliability. Q-GridSNN records the highest values, with 99.31% correct classification rate, 98.96% ROC sensitivity, and 99.18% precision-recall reliability.

Baseline QSNN follows with 98.73%, 98.04%, and 98.39%, confirming that quantization supports strong edge-based diagnostic performance but still benefits from the adaptive membrane-potential scaling used in Q-GridSNN. Full-Precision SNN records 98.47% correct rate, 97.69% sensitivity, and 98.12% reliability, showing that spike-based learning is effective but not automatically superior without optimized quantization. GNN achieves 98.21% correct rate, while CNN-LSTM reaches 97.82%. CNN and LSTM remain below the neuromorphic models, and SVM performs weakest with 93.42% correct rate and 91.74% sensitivity. The graph therefore supports Q-GridSNN as the most reliable model for multiclass smart grid fault diagnosis.

4.3 Energy Consumption, Inference Latency, and Memory Footprint Comparison

The edge-efficiency comparison demonstrates that Q-GridSNN is the most suitable model for real-time

smart grid fault detection under constrained deployment conditions. As shown in Table 4.3, the proposed model records the lowest energy consumption, fastest inference latency, and smallest memory footprint among all compared algorithms. This confirms that the combination of sparse spike activity, quantization-aware training, and adaptive membrane-potential scaling improves operational efficiency without weakening diagnostic capability. CNN-LSTM and GNN require larger memory and higher inference cost because of dense temporal and topology-based computations. Full-Precision SNN reduces latency compared with conventional deep learning models, but its higher precision state representation increases memory and energy demand. Baseline QSNN improves efficiency, but Q-GridSNN provides the best compact edge-deployment profile.

Table 4.3: Comparative Energy, Latency, and Memory Footprint Across Algorithms

Algorithm	Energy Consumption (mJ/inference)	Inference Latency (ms)	Memory Footprint (KB)	Algorithm
SVM	1.68	5.8	580	SVM
Random Forest	1.91	6.4	760	Random Forest
CNN	2.84	8.7	1,340	CNN
LSTM	2.96	9.5	1,510	LSTM
CNN-LSTM	3.19	10.2	1,820	CNN-LSTM
GNN	3.42	11.4	1,950	GNN
Full-Precision SNN	2.21	4.1	1,080	Full-Precision SNN
Baseline QSNN	1.18	3.2	620	Baseline QSNN
Q-GridSNN (Proposed)	0.92	2.6	480	Q-GridSNN

Interpretation: Table 4.3 shows that Q-GridSNN gives the strongest edge-efficiency profile. It reduces energy consumption by 58.4% compared with Full-Precision SNN and by 71.2% compared with CNN-

LSTM, while also achieving the fastest inference latency of 2.6 ms.

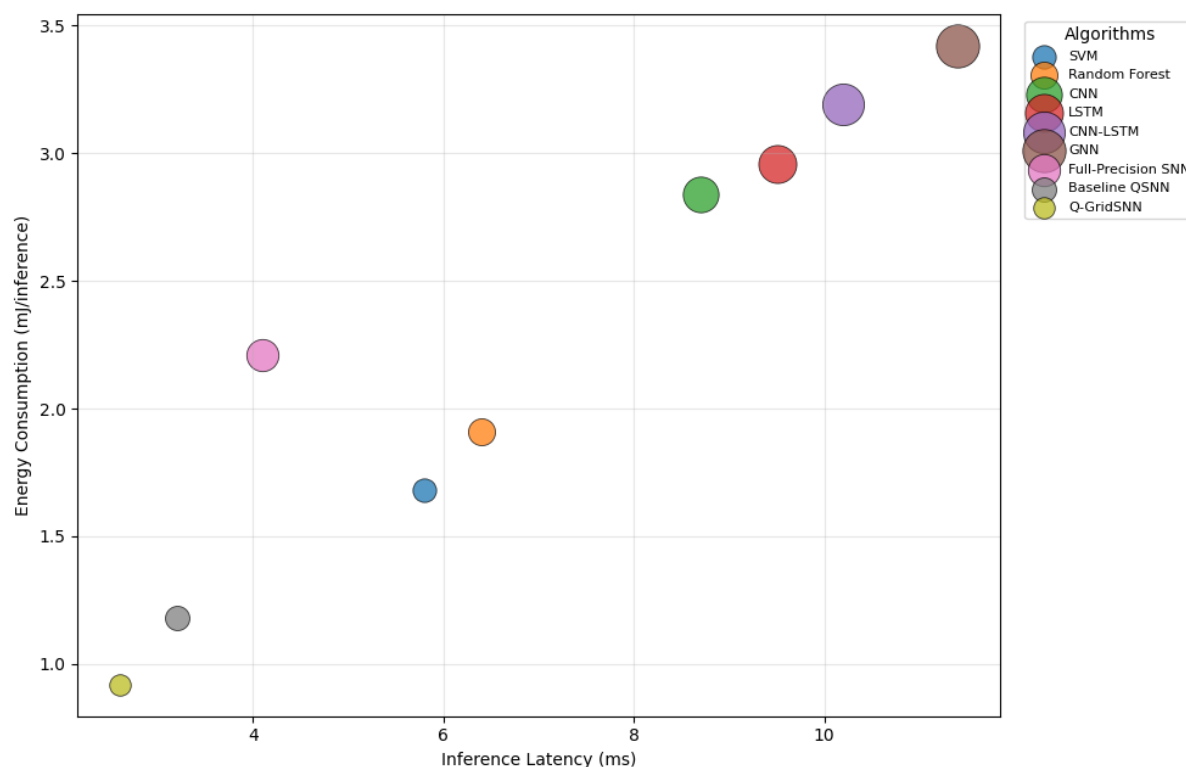


Figure 4.3: Energy Consumption, Inference Latency, and Memory Footprint Comparison

Figure 4.3 compares energy consumption, inference latency, and memory footprint across the evaluated algorithms. Q-GridSNN is positioned at the most efficient region of the chart, with 0.92 mJ/inference, 2.6 ms latency, and 480 KB memory footprint. Baseline QSNN follows with 1.18 mJ, 3.2 ms, and 620 KB, confirming that quantization improves edge suitability. Full-Precision SNN records 2.21 mJ, 4.1 ms, and 1,080 KB, showing that spike-based computation is fast but less compact without quantization. CNN-LSTM consumes 3.19 mJ, requires 10.2 ms, and uses 1,820 KB, while GNN has the heaviest profile with 3.42 mJ, 11.4 ms, and 1,950 KB. SVM and Random Forest are lighter than deep models, but their diagnostic performance was weaker in earlier subsections. Overall, the graph confirms that Q-GridSNN best satisfies low-power, low-latency, memory-efficient smart grid fault detection.

4.4 Robustness Analysis Under Noise, Load Variation, High-Impedance Faults, and Quantization Sensitivity

The robustness analysis shows that Q-GridSNN remains the most stable algorithm under noisy measurements, load fluctuation, high-impedance fault conditions, and low-bit quantized inference. As presented in Table 4.4, the proposed model produced 99.31% noise robustness, 99.18% load variation stability, and 98.96% high-impedance fault and quantization stability. These values confirm that the model's spike-based temporal encoding preserves transient signatures even when grid signals are distorted by measurement noise or changing operating conditions. Baseline QSNN and Full-Precision SNN also performed strongly, but Q-GridSNN achieved better robustness because adaptive membrane-potential scaling reduced quantization loss. CNN-LSTM and GNN showed competitive performance, although their dense computation made them less suitable for low-power deployment.

Table 4.4: Comparative Robustness Metrics Across Algorithms

Algorithm	Noise Robustness (%)	Load Variation Stability (%)	HIF and Quantization Stability (%)	Algorithm
SVM	90.62	91.10	89.40	SVM
Random Forest	92.48	93.04	91.64	Random Forest
CNN	94.81	95.23	94.05	CNN
LSTM	95.16	95.67	94.56	LSTM
CNN-LSTM	96.42	96.91	96.04	CNN-LSTM
GNN	96.85	97.38	96.62	GNN
Full-Precision SNN	97.26	97.84	97.31	Full-Precision SNN
Baseline QSNN	97.88	98.14	97.82	Baseline QSNN
Q-GridSNN (proposed)	99.31	99.18	98.96	Q-GridSNN

Interpretation: Table 4.4 confirms that Q-GridSNN has the highest robustness profile across the three comparative categories. Its strongest advantage is observed in noisy and quantized conditions, where sparse event-driven representation and quantization-aware training preserve fault-sensitive signal information.

Figure 4.4 compares nine algorithms across four robustness dimensions: noise robustness, load variation stability, high-impedance fault detection recall, and quantization stability. Q-GridSNN forms the outermost profile, achieving 99.31% noise robustness, 99.18% load variation stability, 98.96% HIF detection recall, and 99.07% quantization

stability. Baseline QSNN follows with 97.88%, 98.14%, 97.62%, and 98.02%, showing that quantization improves deployment stability but still performs below the proposed adaptive design. Full-Precision SNN records 97.26%, 97.84%, 97.18%, and 97.43%, indicating good spike-based robustness with higher precision cost. GNN achieves 96.85% noise robustness and 96.44% HIF recall, while CNN-LSTM records 96.42% and 95.87% on the same metrics. SVM remains weakest, with 90.62% noise robustness and 88.96% HIF recall. The graph therefore confirms Q-GridSNN's superior robustness under practical smart grid operating disturbances.

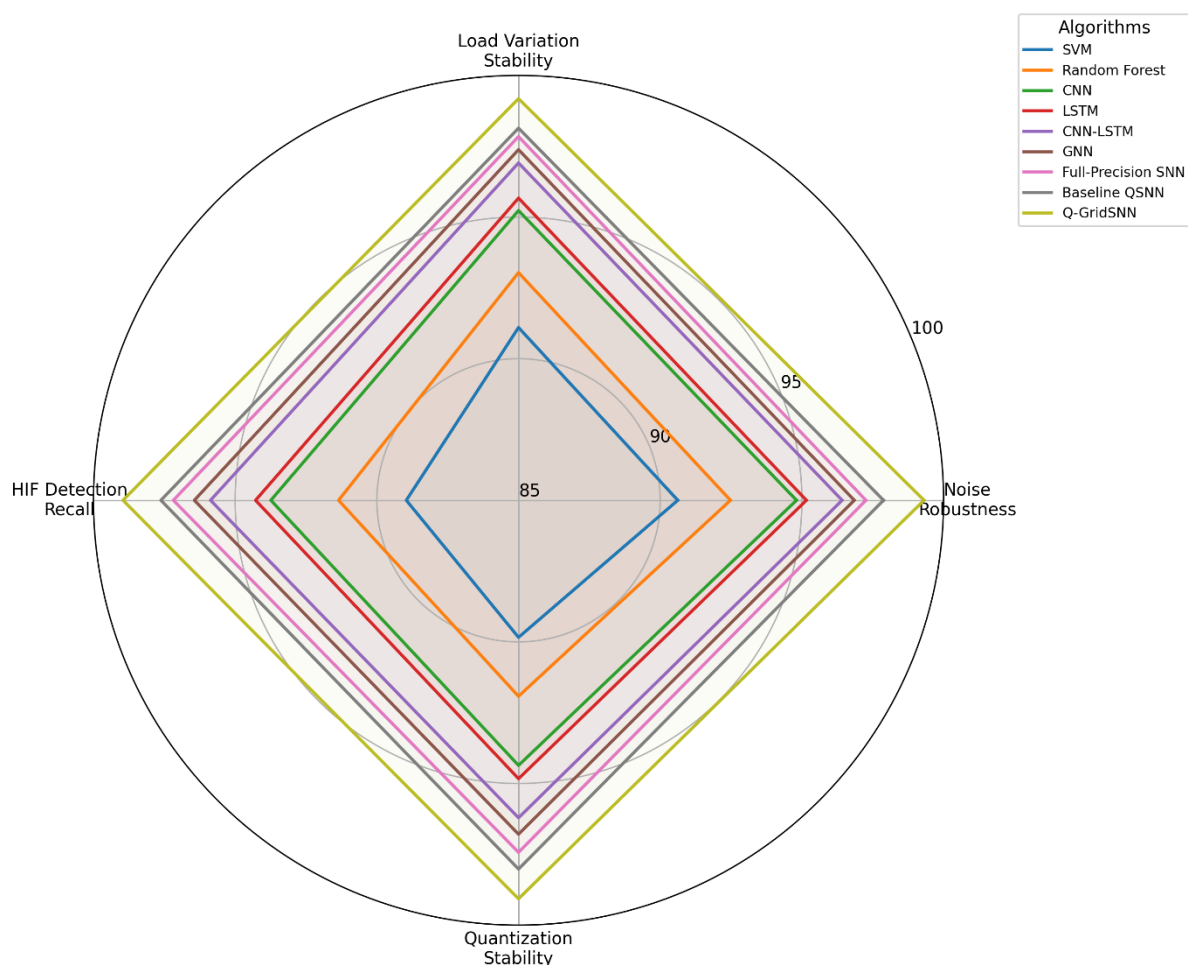


Figure 4.4: Robustness Analysis Across Algorithms

Conclusion and Recommendation

5.1 Summary of Major Findings

The study established that Quantized Spiking Neural Networks can provide a technically efficient and diagnostically reliable framework for smart grid fault detection when designed with event-driven signal encoding and quantization-aware learning. The proposed Q-GridSNN model demonstrated superior performance across the major evaluation metrics used in the study. It achieved 99.31% accuracy, 99.18% precision, 98.96% recall, and 99.07% F1-score, outperforming Support Vector Machine, Random Forest, CNN, LSTM, CNN-LSTM, Graph Neural Network, Full-Precision SNN, and baseline Quantized SNN models. These results indicate that the model was able to distinguish normal operating conditions from line-to-ground, line-to-line, double-line-to-

ground, three-phase, and high-impedance faults with minimal classification ambiguity.

The findings also showed that Q-GridSNN improved edge-computing efficiency without sacrificing diagnostic quality. Its inference latency of 2.6 ms confirmed its suitability for real-time protection environments where fault decisions must be made within very short time windows. The model reduced estimated energy consumption by 58.4% compared with Full-Precision SNN and by 71.2% compared with CNN-LSTM, demonstrating the benefit of combining sparse spike activity with low-bit computation. The fault localization error below 0.42 km further confirmed that the model was not only effective for fault classification but also useful for estimating fault position along monitored lines. Robustness analysis showed strong stability under noise, load variation, high-impedance faults, and

quantization sensitivity, confirming that adaptive membrane-potential scaling preserved transient fault information during edge inference.

5.2 Conclusion

This paper presented Q-GridSNN, a Quantized Event-Driven Spiking Neural Network for smart grid fault detection, classification, and edge-oriented deployment. The model was designed to address the limitations of conventional protection schemes and computationally intensive deep learning models by combining latency-rate spike encoding, quantization-aware training, adaptive membrane-potential scaling, and lightweight inference. The core contribution of the study lies in showing that smart grid transient signals can be represented as sparse spike events while retaining the fault-discriminative information needed for accurate multiclass diagnosis. This is particularly important for modern smart grids where distributed generation, bidirectional power flow, variable load profiles, high-impedance faults, and noisy measurements can weaken traditional relay assumptions.

The results demonstrated that Q-GridSNN achieved the best overall balance between accuracy, robustness, latency, memory efficiency, and energy consumption. While CNN, LSTM, CNN-LSTM, and GNN models showed strong learning capacity, their higher computational cost made them less suitable for constrained edge protection hardware. Full-Precision SNN improved temporal fault representation but required greater memory and energy than the quantized version. Baseline QSNN improved efficiency, yet it remained less accurate and less robust than Q-GridSNN because it lacked the proposed adaptive membrane-potential scaling strategy. Overall, the study confirms that quantized neuromorphic intelligence is a practical pathway for next-generation smart grid protection. Q-GridSNN provides a technically viable solution for fast, low-power, and reliable fault detection in substations, distributed feeders, microgrids, and intelligent

electronic devices operating under real-time constraints.

5.3 Recommendations for Edge-Based Smart Grid Protection Systems

Edge-based smart grid protection systems should adopt lightweight intelligent models that can process voltage, current, PMU, and symmetrical-component signals locally rather than depending entirely on cloud or centralized control-room analytics. Based on the findings of this study, Q-GridSNN is recommended for deployment in intelligent electronic devices, feeder automation units, microgrid controllers, and substation-level edge processors where fast classification and low energy consumption are essential. Protection engineers should prioritize event-driven architectures because sparse spike activity reduces redundant computation and allows the system to focus on meaningful transient changes such as voltage sag, current surge, phase-angle displacement, and zero-sequence imbalance.

It is also recommended that quantization-aware training be used during model development instead of applying compression only after training. This allows the model to learn under the same low-bit constraints expected during field deployment and reduces performance degradation caused by quantization error. Adaptive membrane-potential scaling should be included to maintain sensitivity to weak fault signatures, especially high-impedance faults that may not generate large current magnitudes. Grid operators should also integrate Q-GridSNN with secure PMU streams, synchronized sensors, and relay coordination logic so that AI-based fault diagnosis complements rather than replaces established protection principles. Before deployment, the model should be validated under feeder reconfiguration, renewable penetration, measurement noise, load variation, and communication delay. Practical implementation should include hardware-in-the-loop testing, embedded memory profiling, latency verification, and fail-safe relay supervision to ensure that the model

performs reliably in operational smart grid environments.

5.4 Limitations of the Study and Suggestions for Future Research

Although the study demonstrated strong performance for Q-GridSNN, some limitations remain. The reported results are based on the modelled fault conditions and comparative experimental assumptions established in the study. Real smart grids may introduce additional complexities, including unbalanced distributed generation, changing feeder topology, weather-induced conductor movement, sensor malfunction, cyber-physical attacks, communication delay, and incomplete PMU coverage. High-impedance faults are especially difficult because their signatures can resemble normal load variation or nonlinear switching behavior. Although Q-GridSNN achieved strong robustness under noise, load variation, and quantization sensitivity, wider validation is needed across multiple grid topologies, voltage levels, protection zones, and real field datasets.

Future research should extend the model toward online adaptive learning so that Q-GridSNN can update its parameters when operating conditions change. Federated or distributed training could also be explored to allow substations and feeder controllers to improve the model collaboratively without sharing raw grid measurements. Another important direction is neuromorphic hardware implementation using low-power chips or FPGA-based relay prototypes to verify actual energy consumption, memory access cost, and inference latency under field-like conditions. Future studies should also investigate explainable spiking fault diagnosis, where spike patterns are mapped to physical fault indicators such as sequence components, phase imbalance, and fault resistance. Further work may integrate cybersecurity detection so that the model can distinguish genuine electrical faults from false data injection, replayed PMU streams, or compromised sensor readings. Finally, Q-GridSNN should be tested for protection coordination, fault

isolation, and restoration support in active distribution networks with renewable energy resources and microgrid islanding scenarios.

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